

EthioSchool SaaS

Next-Generation Multi-Tenant School Operating System for Ethiopia



Target Audience: Ethiopian Secondary & Preparatory Schools (Grades 9–12)

Core Calendar: Native Ethiopian Ge'ez (13 Months) & Gregorian Sync

Architecture: Multi-Tenant Database Isolation (TenantAwareModel)

Production Status: Verified Ready (68/68 E2E Tests Passed)

Billing Engine: Automated Seat-Based Metering & Chapa Gateway

Security: Role-Based Access (7 Roles) & QR Cryptographic Verify

1. Executive Summary & Core Value Proposition

EthioSchool SaaS is an enterprise-grade, cloud-native educational management platform engineered specifically to satisfy the operational, regulatory, and pedagogical requirements of Ethiopian high schools. Unlike generic Western school software, EthioSchool is natively built with first-class support for the **13-Month Ethiopian Calendar** (Meskerem to Pagume), Ministry of Education secondary curricula (Grades 9 & 10 General Stream, Grades 11 & 12 Natural/Social Science Streams), 5-component continuous assessment pipelines, and integrated seat-based pricing with local payment gateways.

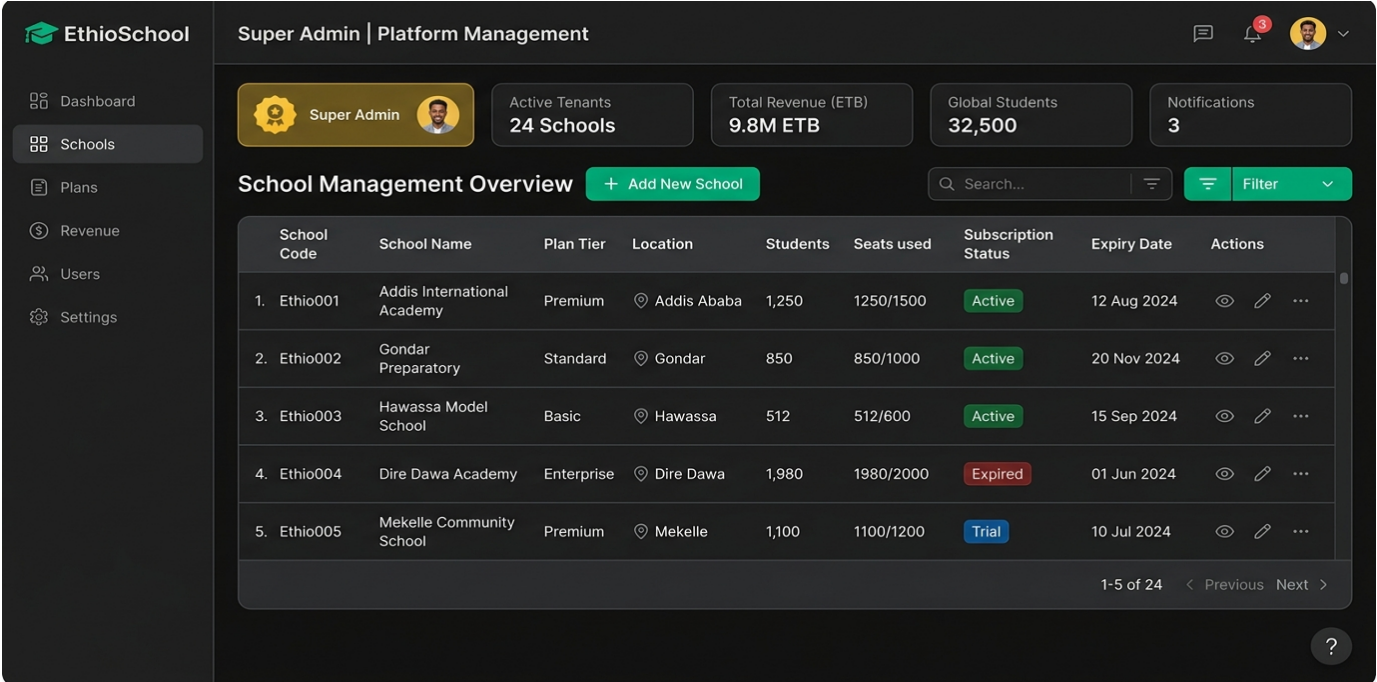
Platform Capability	Technical Architecture & Implementation	Operational Impact
Multi-Tenant Isolation	Strict schema-level query filtering via TenantAwareModel and TenantManager with automated subdomain routing.	100% data privacy between schools; zero risk of cross-tenant data leaks.
Dual-Calendar Engine	Bi-directional algorithmic translation between Ge'ez Ethiopian dates and Gregorian ISO dates across all tables, schedules, and attendance.	Schools operate in native Ethiopian dates while maintaining international compatibility.
Assessment & Ranks	Deterministic 4-phase mark pipeline: <i>Draft</i> → <i>Submitted</i> → <i>Approved</i> → <i>Published</i> , auto-computing 1st–Nth section ranks.	Eliminates manual ranking errors and prevents unauthorized grade tampering.
Automated Rollover	Annual rollover engine: replicates subject schemes, creates promotion histories, advances students, and isolates graduates.	Reduces yearly school transition workload from weeks to a single click.
Tamper-Proof ID Cards	Client/Server PDF ID card generator with signed QR code verification portal and national barcode standards.	Instant mobile verification of student active status by gatekeepers and exam proctors.

2. Role-Based Ecosystem & Permission Matrix

EthioSchool enforces strict least-privilege security across 7 dedicated portal interfaces:

■ Super Admin: Multi-school provisioning, subscription health, revenue analytics, global audit logs.	■ School Admin: Academic calendar setup, staff assignment, fee policies, period closing, rollover.
■■■ Teacher: Daily attendance, continuous assessment mark recording, assignment submission, syllabus.	■ Student: Live gradebook view, downloadable report cards, timetable, exam attempts, homework.
■■■■■ Parent: Multi-child academic progress, real-time attendance alerts, Chapa tuition payment.	■ Registrar & Accountant: ID generation, student admissions, fee invoice issuance, receipts.

3. Super Administrator Platform & Subscription Management



- 1

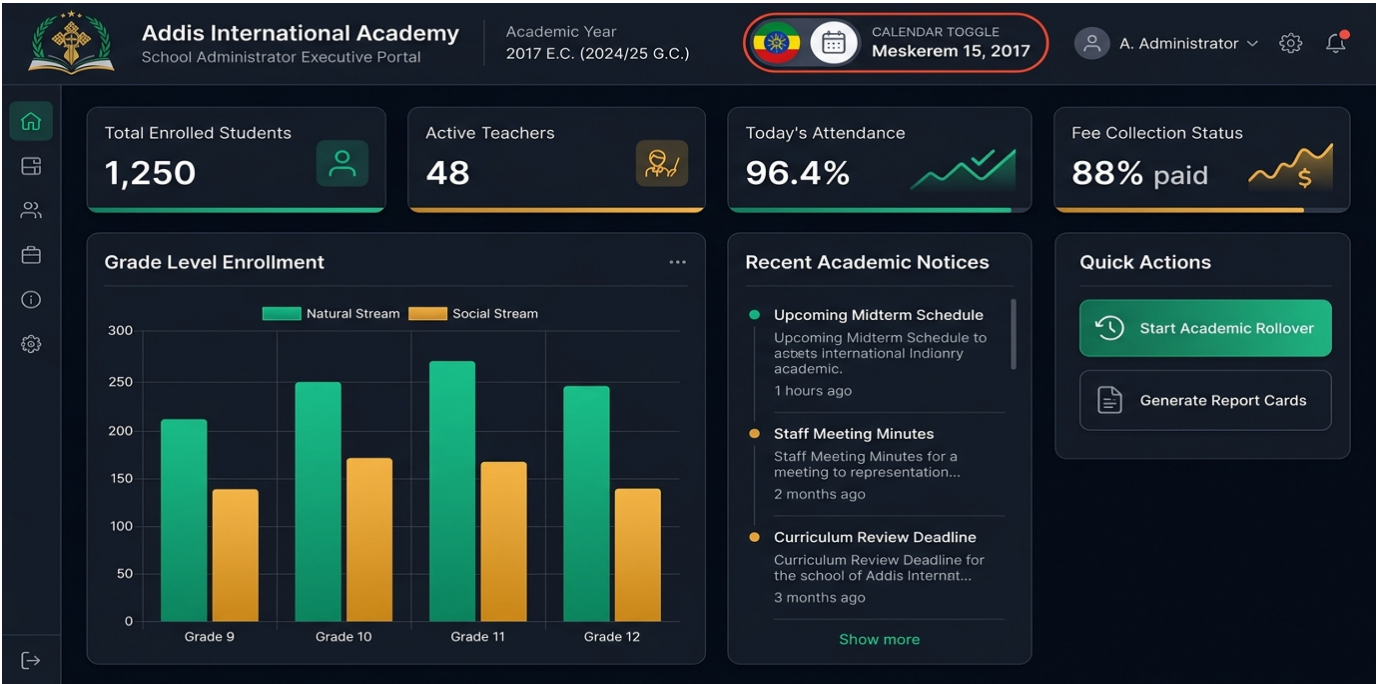
Multi-Tenant Dashboard: Live monitoring of 24+ onboarded schools with aggregated revenue, active student seat counts, and global infrastructure health.
- 2

Seat-Based Usage Metering: Transparent tracking of active student accounts and staff allocations against plan quotas (e.g. 1,250/1,500 seats).
- 3

Dynamic Expiration & Lock Engine: Automated status synchronization marking expired schools with red badges and restricting operational access.
- 4

Tenant Provisioning & Overrides: 1-Click school provisioning with temporary credentials, forced password reset, and custom seat pricing overrides.

4. School Administrator Executive Command Center



- 1

Native Dual-Calendar Toggle: Header badge displays current Ethiopian date (*Meskerem 15, 2017 E.C.*) with a persistent switch to Gregorian dates.
- 2

Live Institutional KPIs: Real-time counters for enrolled students (1,250), active faculty (48), daily attendance (96.4%), and fee collection (88%).
- 3

Stream Breakdown Analytics: Grade 9 to 12 distribution visualizing General Stream vs Natural Science and Social Science enrollments.
- 4

Academic Rollover & Actions: Quick action buttons for executing year-end rollover, scheme replication, and batch report card generation.

5. Teacher Gradebook & Continuous Assessment Matrix



ETHIOPIAN
HIGH SCHOOL

Teacher Portal / Gradebook / Grade 11 / Natural Science / Physics

Search



Dashboard

Gradebook

Attendance



✓ Submit Marks to Admin

Export Excel

Grade Policy

Total Weight: 100% Valid

GRADE 11 PHYSICS (NATURAL SCIENCE) - MARK ENTRY & GRADEBOOK

Section: A

Term: Semester 1

Session: 2023/2024

Status: Approved

	STUDENT ID	STUDENT FULL NAME	CLASS ACTIVITY (10)	QUIZ & HW (10)	PROJECT WORK (10)	MIDTERM EXAM (20)	FINAL EXAM (50)	TOTAL (100)	RANK	STATUS
1.	AIA/11N/001	 Abebe Bekele	9	8	9	17	45	88	2	Approved
2.	AIA/11N/002	 Meron Hailu	10	10	9	19	49	97	1	Approved
3.	AIA/11N/003	 Dawit Tadesse	8	7	8	15	42	80	4	Submitted
4.	AIA/11N/004	 Tigist Lemma	9	9	10	18	46	92	3	Approved
5.	AIA/11N/005	 Yosef Alemu	7	6	8	14	38	73	5	Submitted

Showing 1-10 of 48 Students

Grade 11 | Physics | Semester 1 Marks

1

Ethiopian 5-Component Scheme: Standard Ministry configuration: *Class Activity (10%), Quiz/HW (10%), Project (10%), Midterm (20%), Final Exam (50%) = 100% Total.*

2

Weight Integrity Validation: Real-time weight validation badge (*Total Weight: 100% Valid*) preventing over/under-allocated mark components.

3

4-Phase Approval Workflow: Teacher enters DRAFT → submits to School Admin → Admin APPROVES → marks are PUBLISHED and locked from edits.

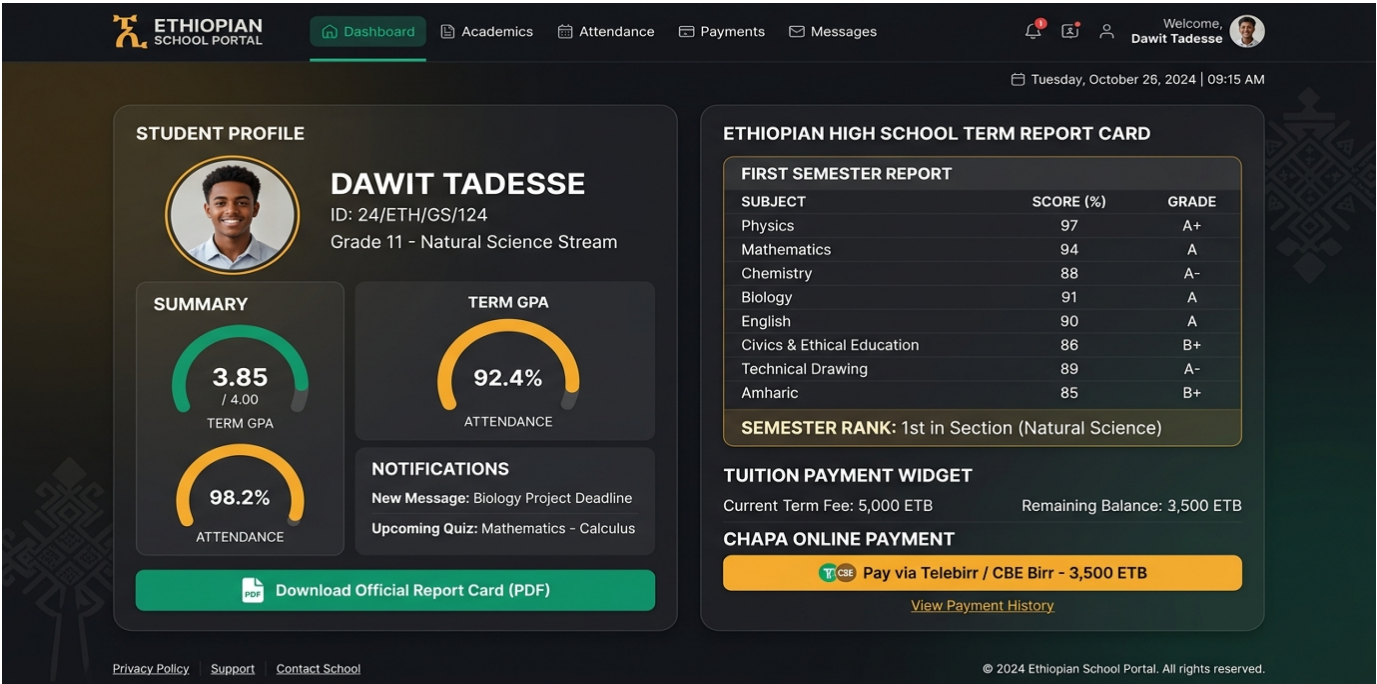
4

Automated Section Rank Computation: Instant deterministic ranking (1st to 48th) calculated across weighted assessment totals.

Ethiopian Secondary & Preparatory School Cloud Solution (Grades 9–12)

Page 5 of 8

6. Student & Parent Portals with Chapa Payments



- 1 Student Academic Profile & Performance Gauges:** Real-time GPA gauge (3.85 / 4.00), attendance percentage (98.2%), and notification feed.
- 2 Ministry-Compliant Report Card:** Complete subject performance breakdown with grades (A+, A, A-, B+), semester GPA, and official section rank.
- 3 Chapa Online Tuition Payment:** 1-Click online fee payment supporting Telebirr, CBE Birr, and debit cards with automated receipt generation.
- 4 Digital PDF Download:** Instant download of official, digitally stamped student report cards for parents and university admissions.

7. Tamper-Proof Student ID Cards & QR Verification



ADDIS INTERNATIONAL ACADEMY

ID Generator

QR Verification

Manage Students

Reports

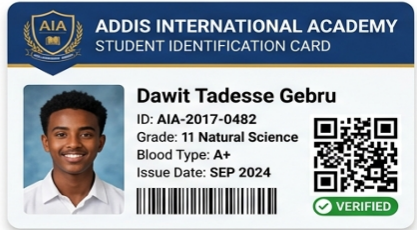
Account (Admin)

STUDENT ID CARD GENERATOR

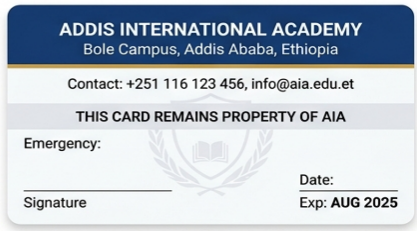
Generate ID

Download PDF

Front



Back



DIGITAL QR VERIFICATION PORTAL

New Scan

SCAN QR CODE


OR

 Upload

VERIFICATION RESULTS

Green Header



VERIFIED STUDENT - ACTIVE ENROLLMENT

Student: DAWIT TADESSE GEBRU

ID: AIA-2017-0482

Grade: 11 Natural Science

Status: ACTIVE

Verification Details

Cryptographic Hash: 9a7f3d...e8c1b2967a7dd9ac38Idce23383307...

Verifier ID: ADMIN_MAIN_ENTRANCE

Valid Until: AUG 2025

Verified At: SEP 28, 2024, 09:12 AM

RECENT SCAN HISTORY

Date/Time	Location	Student	Status
SEP 28, 09:12 AM	Main Entrance	Dawit Gebru	Verified
SEP 27, 08:01 AM	Library	Dawit Gebru	Verified
SEP 26, 12:30 PM	Cafeteria	Dawit Gebru	Verified

- 1 Official Dual-Sided PVC ID Layout:** High-resolution card with school crest, photo, English student full name, ID number, blood group, emergency rules, and issue date.
- 2 Public QR Verification Engine:** Anyone scanning the card QR code with any smartphone is taken to a secure verification portal confirming active enrollment.
- 3 Cryptographic Tamper-Proof Signature:** Digital signature hash prevents student ID forging or expired ID reuse.
- 4 Batch PDF Generation:** Registrars can generate print-ready PDF sheets for entire grade levels or single replacement cards instantly.

8. Production Verification & 2-Year Lifecycle Audit

To guarantee 100% production readiness, the system underwent a rigorous, automated **2-Year School Lifecycle Simulation** covering multi-stream enrollments, continuous assessment pipelines, period closing, student promotion, assessment scheme replication, and multi-tenant security isolation. Below is the verified audit summary:

Verification Stage	Tested Scope & Invariants	Result
1. School & Stream Provisioning	Creation of 6 grades (9, 10, 11-NS, 11-SS, 12-NS, 12-SS), 30 subjects, 6 faculty, 60 enrolled students.	68 / 68 PASSED 100% Verified
2. Mark Recording Pipeline	1,500 continuous assessment marks recorded, submitted, approved, and published with weight integrity validation.	PASSED Admin guard enforced
3. Period Closing & Ranks	Automated GPA calculation and unique section ranking (1st to 10th per class) across 60 period result records.	PASSED Ranks 100% unique
4. Year-End Promotion & Graduation	Annual result aggregation: 40 students promoted (Grades 9–11) and 20 Grade 12 students verified as GRADUATED.	PASSED Zero orphan enrollments
5. Year 2 Scheme Replication	Replicated 150 assessment components from AY1 to AY2; verified exactly 100% weight per subject with zero weight accumulation bugs.	PASSED Weights strictly 100%
6. Multi-Tenant Security	Cross-tenant intrusion test: verified attacker school cannot view, modify, or query marks, students, or enrollments.	PASSED Full Tenant Isolation

Enterprise Deployment & Licensing Contact:

- **Platform:** EthioSchool SaaS Enterprise Cloud Edition
- **Primary Phone:** +251 937 381 801
- **Email:** support@ethioschool.edu.et | info@addisacademy.edu.et
- **Hosting & SLA:** Dedicated PostgreSQL Tenant DBs, Daily Backups, 99.9% Uptime Guarantee